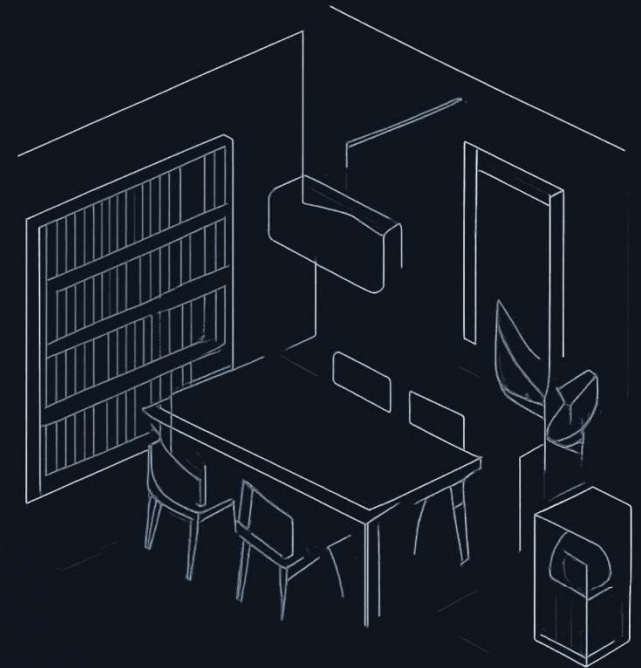


AI-Enabled Smart Library Management System

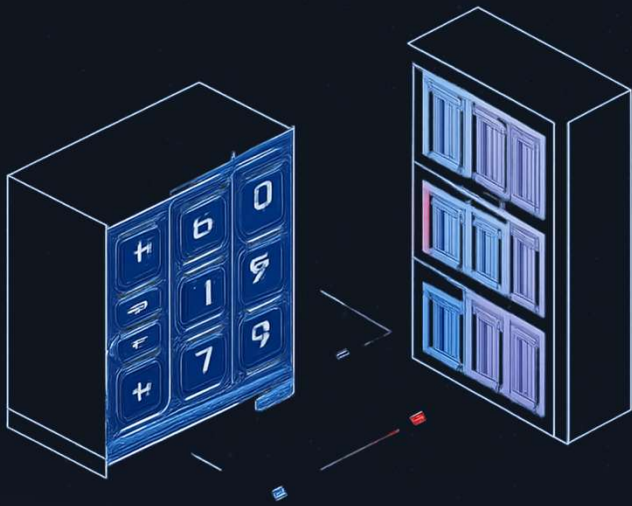
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Why current systems fall short



- Search is slow and imprecise — keyword-only, limited context
- Manual tracking of loans and due dates causes delays and errors
- Notifications are generic or absent, increasing late returns
- Administrative workflows are repetitive and time-consuming

Project Objectives



Intelligent Search

Context-aware search and conversational assistance for students



Efficient Management

Streamlined handling of books, loans, fines, and student records



Automated Reminders

Smart due-date alerts, renewals, and overdue notices via multiple channels



AI Analytics & Recommendations

Personalized reading suggestions and usage insights for librarians

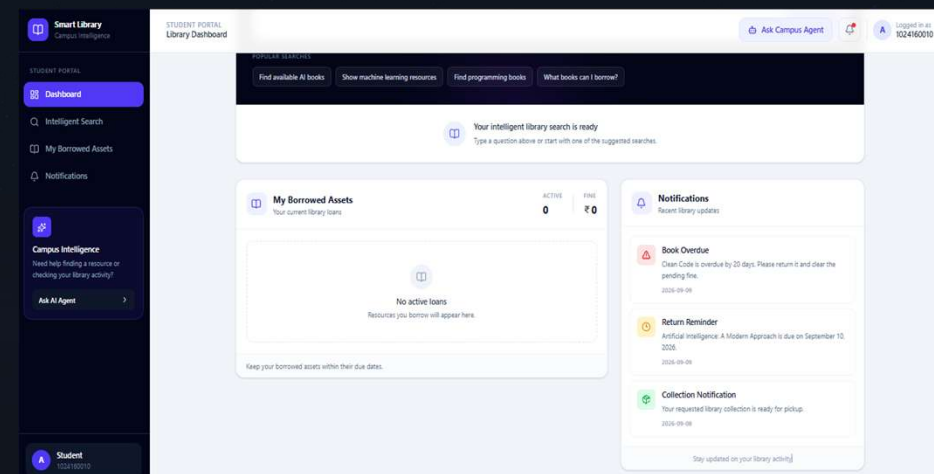
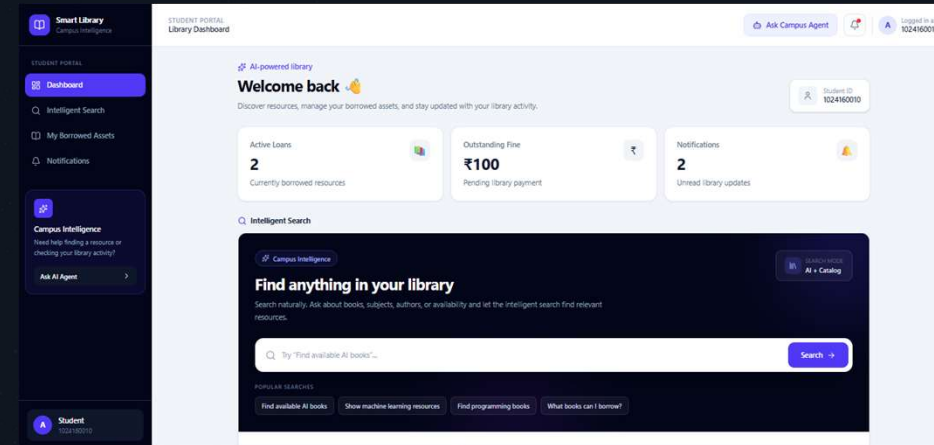
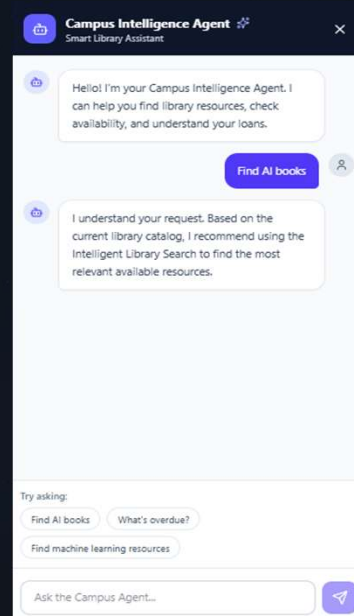
Methodology — System Workflow



Key AI components: • Campus Intelligence Agent — campus-wide context and catalog indexing. • Student Onboarding Agent — profile setup, preferences, and access control. • Smart Monitoring & Notification Agent — due-date tracking and alerts.

Implementation & Early Results

- Use case diagrams and DFDs designed
- React frontend prototype built — student dashboard ready
- Features: Search, Loan Tracking, Notifications, AI Assistance
- Prototype demo uses mock data; SQL schema under review
- Documentation (MkDocs) published via GitHub Pages



Conclusion & Next Steps

Conclusion

Prototype and architecture foundations are complete. Frontend and DB foundations are in place for integration with backend and AI agents.

Next Steps

- 1) Connect frontend to SQL database
- 2) Implement backend APIs and authentication
- 3) Integrate and test AI agents (search, monitoring, recommendations)
- 4) End-to-end testing and user evaluation in campus environment